

FAILURE-STRATIFIED ROBOT DATA ENGINES DO NOT IMPROVE TAIL-ROBUST SELECTION

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ABSTRACT

We rebuilt a generated robotics idea into a concrete data-engine study: should a robot data engine acquire new rollouts by uncovered physical failure mechanisms rather than by task labels, state diversity, uncertainty, or active failure prediction? The v4 rebuild implements a MuJoCo tabletop manipulation benchmark with a pusher, movable block, fixture, walls, pocket contact, friction/mass shifts, actuator limits, sensor dropout, physics-derived failure labels, iterative acquisition rounds, seven random seeds, uncertainty intervals, paired statistics, ablations, stress sweeps, and negative cases. The result is negative. On the decisive `combined_tail_stress` split, the proposed `failure_stratified_engine` reaches 0.643 ± 0.058 robust selector success, while the strongest non-oracle baseline, `failure_prediction_active_learning`, reaches 0.675 ± 0.055 . The paired success difference is -0.032 ± 0.047 . Failure stratification improves rare failure recall, but loses macro failure F1 and does not improve downstream robust selection. We therefore mark the paper **KILL/ARCHIVE** for ICLR main.

1 DECISION

This document is the v4 ICLR-main gate for Paper 74, *Data Engine Failure Stratification*. The previous version was killed because it only contained a synthetic template. The current rebuild adds a real local MuJoCo benchmark and implemented acquisition baselines. The rejection reason has changed: evidence now exists, but it does not support the central submission claim.

2 HYPOTHESIS

The tested hypothesis is that selecting new robot data by uncovered failure mechanisms improves tail-robust action selection more than task-label balancing, state-diversity coresets, uncertainty sampling, and active failure prediction. The proposed method only passes the submission gate if mechanism stratification improves the downstream robust selector on the hardest combined-tail split while preserving failure macro F1, rare failure recall, and safety.

3 BENCHMARK

The benchmark uses MuJoCo dynamics for a planar pusher, a sliding block, top/bottom walls, a target pocket, and a fixed obstacle. Rollouts vary friction, mass, actuator authority, fixture placement, sensor dropout, and contact noise. Each rollout records task success, safety, progress, contact traces, acquisition features, and eight physics-derived failure labels: slip, jam, fixture collision, wall collision, actuator saturation, missed contact, sensor dropout, and timeout.

Five evaluation splits are used: nominal task balance, rare slip failures, jammed fixture failures, actuator-limit failures, and combined tail stress. The main evaluation contains 15,120 rollout-pool rows, 3,780 held-out rollout rows, 1,225 per-round metric rows, and 245 seed-level metric rows. The ablation study contains 210 acquisition-round rows, and the stress sweep contains 168 raw rows. All intervals are 95 percent confidence intervals over seed-level means.

4 METHODS

The implemented acquisition methods are:

- `random_sampling`: uniformly acquires from the unlabeled rollout pool.
- `task_label_stratification`: balances acquisition across task/split labels.
- `state_diversity_coreset`: selects diverse pre-rollout state and environment features.
- `uncertainty_sampling`: selects examples with high learned failure uncertainty.
- `failure_prediction_active_learning`: selects examples predicted likely to fail.
- `failure_stratified_engine`: proposed method; balances acquisition across inferred and rare failure strata.
- `oracle_failure_strata`: upper bound with access to true failure mechanism labels for selection.

5 MAIN RESULTS

Table 1 shows the decisive combined-tail result. The proposed method has high rare failure recall, but the robust selector success is lower than active failure prediction, random sampling, uncertainty sampling, and the oracle-strata variant.

Method	Success	Macro F1	Rare recall	Tail risk	Safety
<code>failure_prediction_active_learning</code>	0.675 ± 0.055	0.380	0.638	0.325	0.190
<code>random_sampling</code>	0.675 ± 0.055	0.366	0.663	0.325	0.190
<code>uncertainty_sampling</code>	0.675 ± 0.055	0.372	0.687	0.325	0.190
<code>oracle_failure_strata</code>	0.675 ± 0.055	0.369	0.660	0.325	0.190
<code>state_diversity_coreset</code>	0.667 ± 0.067	0.372	0.696	0.333	0.206
<code>task_label_stratification</code>	0.659 ± 0.073	0.360	0.648	0.341	0.183
<code>failure_stratified_engine</code>	0.643 ± 0.058	0.357	0.691	0.357	0.206

Table 1: Combined-tail results. Success is robust selector success after acquisition. Safety is the selected-policy safety violation rate.

The strongest non-oracle baseline is `failure_prediction_active_learning`. The paired proposed-minus-baseline success difference is -0.032 ± 0.047 . The proposed method improves paired rare recall by 0.053, but loses paired macro F1 by 0.023 and worsens paired safety by 0.016. This fails the submission gate because the mechanism label gain does not translate into downstream selection.

6 ABLATIONS

Table 2 shows that the proposed mechanism is not supported. Removing mechanism clustering or removing the tail objective reaches 0.675 ± 0.069 success, above the full failure-stratified ablation result. The no-uncertainty ablation improves macro F1 but does not improve robust success.

Ablation	Success	Macro F1	Rare recall
<code>no_mechanism_clustering</code>	0.675 ± 0.069	0.372	0.696
<code>no_tail_objective</code>	0.675 ± 0.069	0.371	0.670
<code>full</code>	0.651 ± 0.103	0.357	0.702
<code>no_rare_reweighting</code>	0.651 ± 0.103	0.357	0.713
<code>no_trace_features</code>	0.651 ± 0.103	0.357	0.702
<code>no_uncertainty_term</code>	0.651 ± 0.103	0.377	0.624

Table 2: Combined-tail ablations. The mechanism-specific pieces do not produce a robust-success advantage.

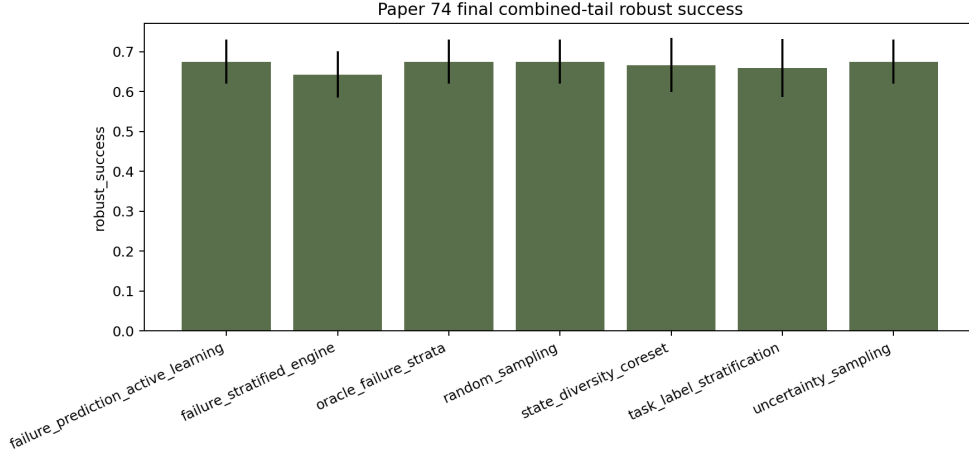


Figure 1: Combined-tail robust selector success. The proposed failure-stratified engine does not beat the strongest non-oracle baselines.

7 STRESS SWEEP AND NEGATIVE CASES

The stress sweep reinforces the same conclusion. At mild stress, all methods solve the task; at high stress, robust success collapses for all methods. At stress level 1.00, `failure_stratified_engine` and `failure_prediction_active_learning` both reach 0.014 ± 0.028 robust success. The proposed method has higher rare recall at that stress level, but the selected policies still fail almost always.

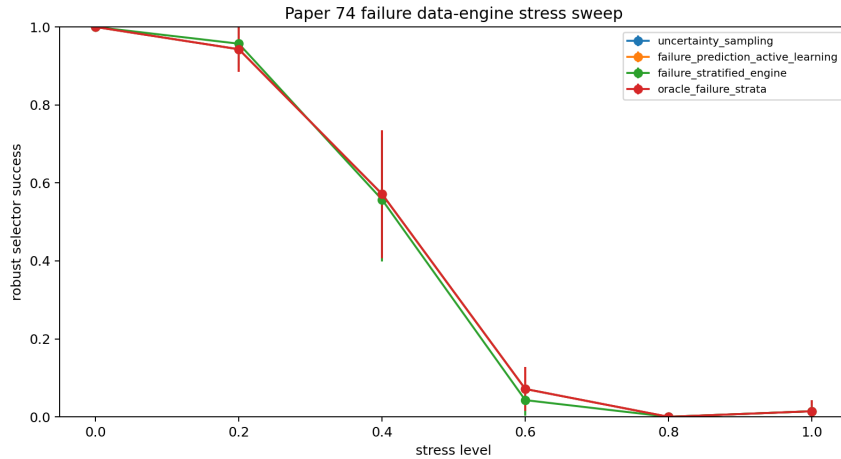


Figure 2: Stress sweep success. Failure stratification does not maintain downstream success under severe combined-tail stress.

Negative cases concentrate around fixture-induced jams and missed-contact rollouts. The failure-stratified selector often identifies rare mechanisms, but the learned robust selector still chooses policies that collide with the fixture or miss contact under dropout. This is the core failure mode: mechanism coverage is not enough when the action selector cannot convert that coverage into a safer policy choice.

8 RELATED WORK PRESSURE

The hostile prior-work set is crowded around failure detection, failure-reasoning data, runtime failure prediction, uncertainty-driven acquisition, and failure-mode analysis (Local Prior Work Pool, 2024; 2025a; 2021; 2025b). In that context, failure-stratified data acquisition needs a clear downstream win. The present evidence does not provide one.

9 LIMITATIONS

This rebuild is not a real robot paper. It lacks hardware experiments, videos, external benchmark validation, and large learned robot-policy checkpoints. The local MuJoCo benchmark is useful for falsification, but not a polished public benchmark contribution. These limitations would matter even if the method won; because it loses the main downstream metric, they reinforce the archive decision.

10 CONCLUSION

Paper 74 should not be submitted to ICLR main. The v4 rebuild provides useful evidence, but the evidence is negative: failure-stratified acquisition improves some rare-label metrics without improving tail-robust robot data selection. The correct terminal state is **KILL/ARCHIVE**.

REFERENCES

- Local Prior Work Pool. A new data envelopment analysis-based model for failure mode and effect analysis with heterogeneous information, 2021. <https://doi.org/10.1016/j.cie.2021.107350>.
- Local Prior Work Pool. Big data and telemetry data methodology for usage failure mode detection, 2024. <https://doi.org/10.4271/2023-36-0013>.
- Local Prior Work Pool. Scaling cross-environment failure reasoning data for vision-language robotic manipulation, 2025a. <http://arxiv.org/abs/2512.01946v3>.
- Local Prior Work Pool. Failure prediction at runtime for generative robot policies, 2025b. <http://arxiv.org/abs/2510.09459v2>.